Stage 8 Maths — Practice Test 1

Paper 1: 60 minutes · no calculator · Paper 2: 60 minutes · calculator allowed · Total 100 marks

Answer all questions. Write your answers in the spaces provided. Show your working where there is space for it. The number of marks for each question is shown in brackets [ ].

Name _____ Class _____ Date _____ Mark: ____ / 100

Abdelkriem Academy · practice paper written in the style of Cambridge Checkpoint. Not an official Cambridge paper.

Paper 1 — 60 minutes — 50 marks

1 **[1]**

Question 1

Work out $20 - 3 \times (2 + 4)^2 \div 12$

Working

Answer

2 **[1]**

Question 2

Which of these numbers is an integer but **not** a natural number?

- ☐ A) −7
- ☐ B) 7
- ☐ C) −7/2
- ☐ D) 0.7

3 **[1]**

Question 3

Hassan writes: $3^4 \times 3^2 = 9^6$

Explain Hassan's mistake and write the correct answer as a single power of 3.

4 **[1]**

(a) Work out $\sqrt[3]{(-64)}$

Working

Answer

(b) Work out $(-3)^2 + \sqrt[3]{(-8)}$ **[1]**

Working

Answer

5 **[1]**

(a) Work out 34×0.1

Working

Answer

(b) Work out $2.7 \div 0.01$ **[1]**

Working

Answer

6 **[1]**

Question 6

Which fraction is equal to a **recurring** decimal?

- ☐ A) 3/8
- ☐ B) 7/20
- ☐ C) 5/6
- ☐ D) 9/25

7 **[2]**

Question 7

Work out 4 1/5 − 1 2/3

Show your working. Give your answer as a mixed number in its simplest form.

8 A bakery in Heliopolis makes biscuits and bread. **[1]**

(a) The baker has 84 chocolate biscuits and 120 date biscuits. She puts all of them into bags. Every bag has the same number of chocolate biscuits and the same number of date biscuits, and none are left over.

What is the **greatest** number of bags she can fill?

Working

Answer

(b) How many date biscuits are in each bag? **[1]**

Working

Answer

(c) One batch of bread dough uses 2 3/4 kg of flour. **[2]**
Work out how much flour the baker needs for 6 batches. Show your working.

Working

Answer ____ kg

(d) The baker has 6 kg of sesame seeds. She puts them into bags that each hold 3/8 kg. **[1]**
How many bags can she fill?

Working

Answer ____

(e) She has 13.2 kg of biscuit dough. Each baking tray holds 0.4 kg of dough. **[2]**
How many trays can she fill? Show your working.

Working

Answer ____

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9 **[1]**

Question 9
Work out 0.25 × 37 × 4 without a calculator.
Show how you can change the order to make the calculation easier.

10 **[1]**

Question 10
Which symbol makes this statement true?
−0.65 ☐ −2/3

- ☐ A) <
- ☐ B) >
- ☐ C) =
- ☐ D) ≤

11 Year 8 students from a school in Cairo go on a trip to Luxor. **[1]**

(a) The total cost of the trip is E£48 652.
Round E£48 652 to 2 significant figures.

Working

Answer

(b) The ratio of students to teachers on the trip is 15 : 2. There are 68 people altogether. **[1]**
How many teachers are on the trip?

Working

Answer

(c) The group spends 45 minutes at a museum and 2 hours at the Karnak temples. **[1]**
Write the ratio 45 minutes : 2 hours in its simplest form.

(d) The school shares E£4500 between a boat trip, museum tickets and lunch in the ratio 2 : 3 : 5. **[2]**
How much money is for lunch? Show your working.

Working

Answer

12 **[1]**

Question 12
Which of these is a **formula**?

- ☐ A) 3x + 7
- ☐ B) 2x + 5 = 11
- ☐ C) 4n − 1 > 6
- ☐ D) P = 2l + 2w

13 **[1]**

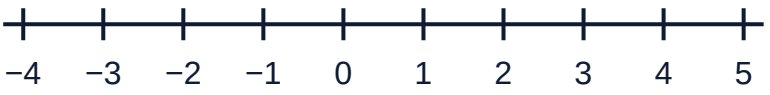
(a) Expand 4x(3x − 2)

(b) Factorise completely $8a^2 + 12a$ **[2]**

14 **[2]**

Question 14
Solve 5x − 7 = 2x + 11
Show your working.

15 **[1]**



Question 15
Show the inequality −2 < x ≤ 3 on the number line. (Copy the number line onto paper.)

16 Here is a sequence: 7, 4, 1, −2, … **[2]**

(a) Find an expression for the nth term of the sequence.

(b) Is −50 a term in this sequence? Show how you decide. **[1]**

17 **[1]**

Question 17

Here is a function machine.

input → × 4 → − 3 → output

The output is 25. Work out the input.

Working

Answer

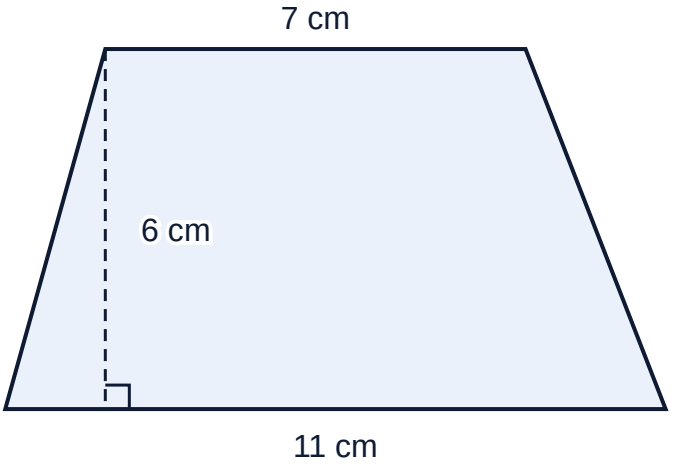
18 **[1]**

Question 18

Nour says, "A rhombus is always a square."

Is Nour correct? Explain your answer.

19 **[2]**



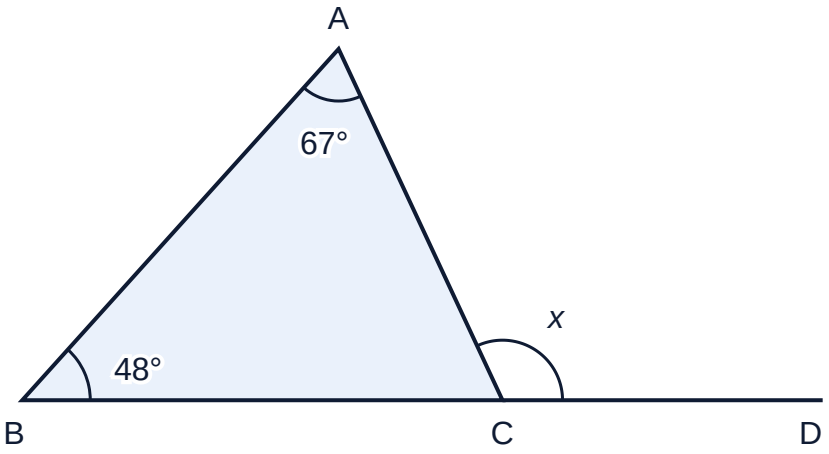
Question 19

Work out the area of this trapezium. Show your working.

Working

Answer ____ cm^2

20 BC is extended to D. **[1]**



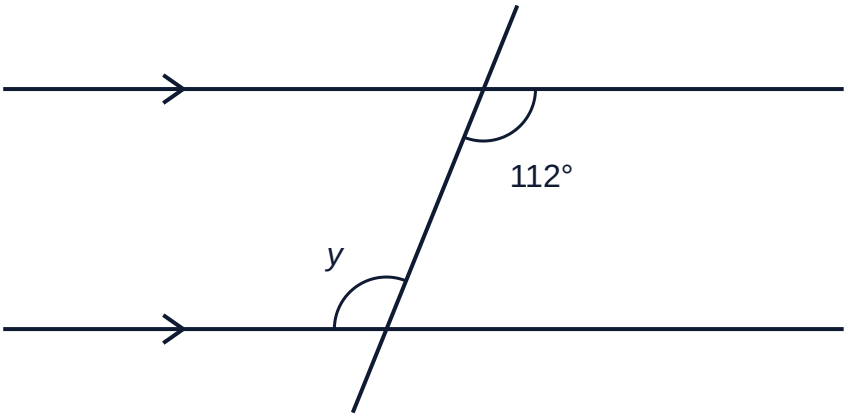
(a) Work out the size of angle x.

Working

Answer ____ °

21 (continued) **[1]**

The diagram shows two parallel lines crossed by a straight line.



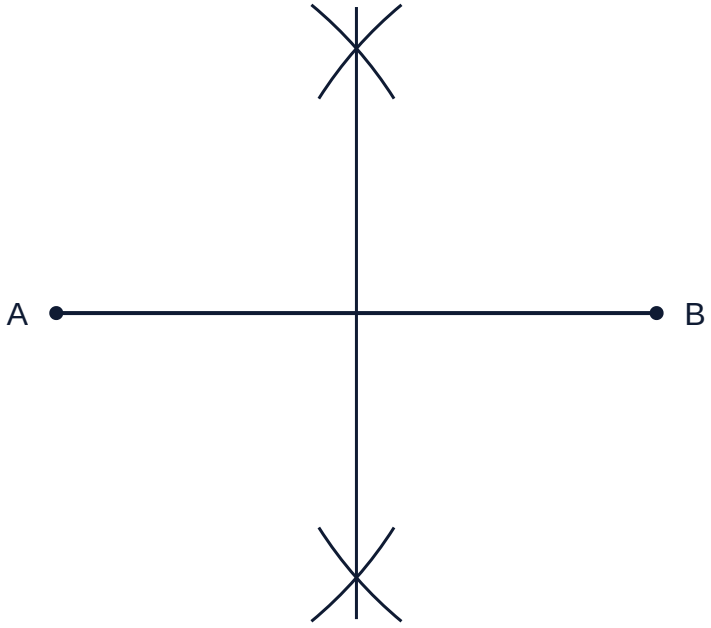
(b) Work out the size of angle y.

Working

Answer °

(c) Give a geometric reason for your answer to part (b). **[1]**

22 **[1]**



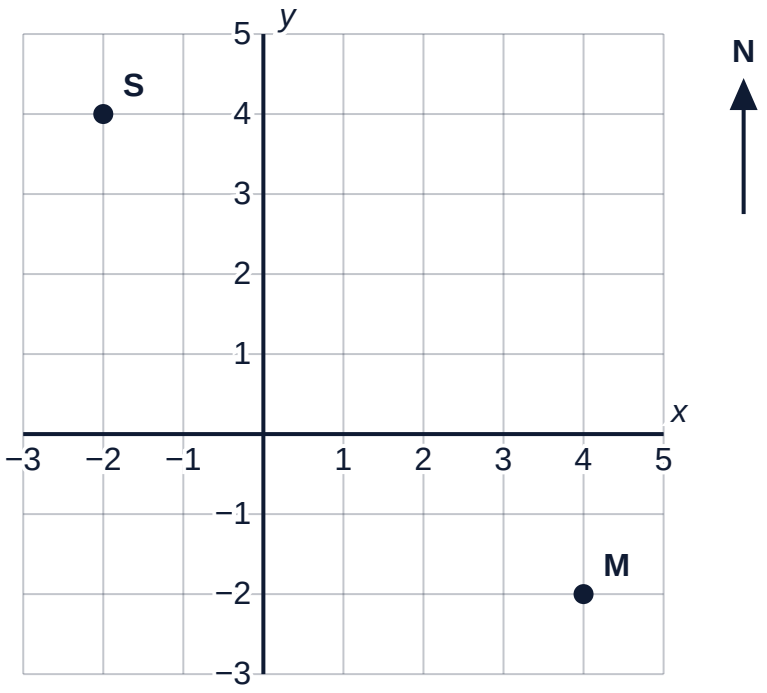
Question 21

Amir uses a ruler and compasses. He draws arcs with the same radius from A and from B, then a straight line through the points where the arcs cross.

What has he constructed?

- ☐ A) The perpendicular bisector of AB
- ☐ B) The bisector of an angle
- ☐ C) A triangle with sides 5 cm, 5 cm and 3 cm
- ☐ D) A line parallel to AB

23 The map of part of a city is drawn on a coordinate grid. S is a bus station and M is a museum. **[1]**



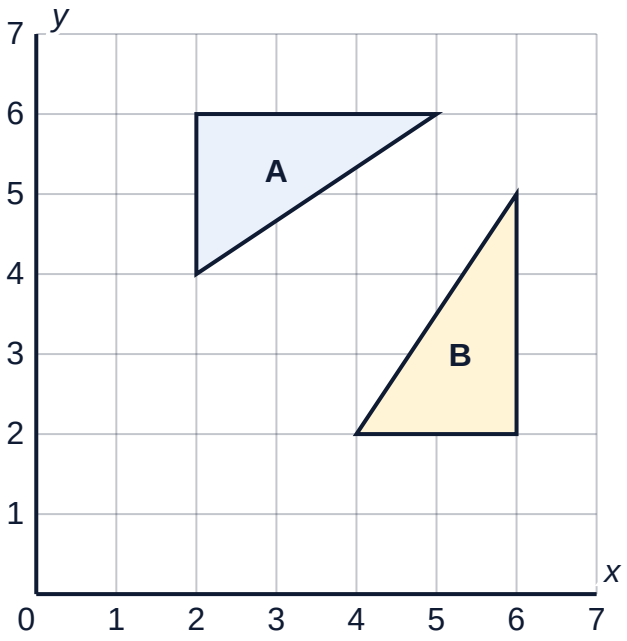
(a) Write down the coordinates of the midpoint of SM.

(b) The bearing of M from S is 135°. **[1]**
Work out the bearing of S from M.

Working

Answer °

24 **[2]**



Question 23

Describe fully the single transformation that maps triangle A onto triangle B.

25 For the science fair, Salma grew 15 bean plants. The stem-and-leaf diagram shows their heights in cm. **[1]**

Stem	Leaf
1	4 7 8
2	0 3 3 6 9
3	1 2 5 8
4	0 2 6
Key: 2	3 means 23 cm

(a) Find the median height.

Working

Answer cm

(b) Find the range of the heights. **[1]**

Working

Answer cm

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26 At the science fair there is a game with spinners. **[1]**

(a) The probability that a spinner lands on red is 0.35.
Work out the probability that it does **not** land on red.

Working

Answer

(b) Another spinner has three equal sections numbered 1, 2 and 3. Ali spins it and also flips a fair coin. **[1]**
Work out the probability that he gets an odd number **and** a head.

Working

Answer

Paper 2 — 60 minutes — 50 marks

1 A mobile-phone shop in Nasr City sells phones and accessories. **[2]**

(a) A phone costs E£14 500. In a sale the price is reduced by 12%.
Work out the sale price. Show your working.

Working

Answer

(b) The price of another phone increased from E£9600 to E£11 040. **[1]**
(i) Work out the absolute change in the price.

Working

Answer

(b)(ii) Work out the percentage increase. Show your working. **[2]**

Working

Answer %

(c) 3 screen protectors cost E£247.50. **[1]**
Work out the cost of 7 screen protectors.

Working

Answer

2 **[1]**

(a) Estimate the value of 487 × 62 ÷ 19
Show your rounding.

(b) Use your calculator to work out 487 × 62 ÷ 19. Give your answer correct to 3 significant figures. **[1]**

Working

Answer

3 Six teams play in a school football tournament. Each team plays every other team once. **[1]**

Team	Won	Drawn	Lost	Goals scored	Goals conceded	Points
Al-Nour	4	1	0	15	4	13
Falcons	3	1	1	9	7	10
Pharaohs	2	2	1	12	8	?
Nile Stars	2	1	2	9	10	7
Delta	1	1	3	7	12	4
Sphinx	0	0	5	5	16	0

The number of points P is given by the formula P = 3w + d, where w is the number of wins and d is the number of draws.

(a) Work out the number of points for the Pharaohs.

Working

Answer

(b) Rearrange the formula P = 3w + d to make w the subject. **[2]**

(c) Work out the mean number of goals scored per team. **[1]**

Working

Answer

(d) Find the median number of goals scored. **[1]**

Working

Answer

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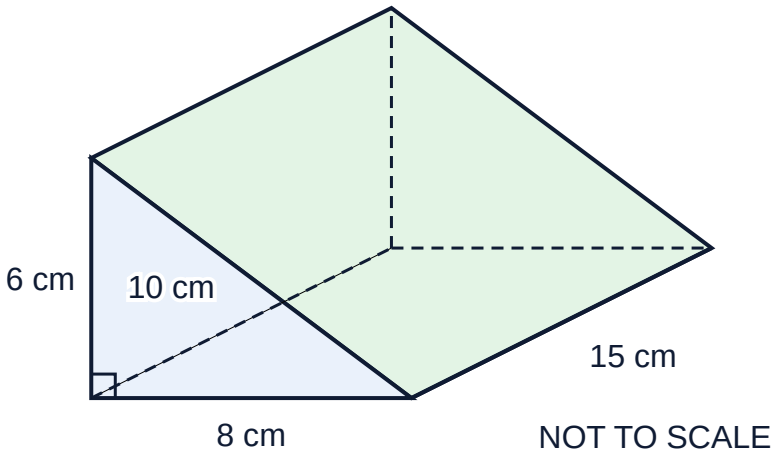
4 The bakery from Paper 1 uses trays and boxes. **[2]**

(a) A round baking tray has a diameter of 38 cm.
Work out the circumference of the tray. Give your answer correct to 1 decimal place.

Working

Answer cm

[2]



(b) A slice of cake is a triangular prism. Its cross-section is a right-angled triangle with sides 6 cm, 8 cm and 10 cm, and the slice is 15 cm long.
Work out the volume of the slice. Show your working.

Working

Answer ____ cm^3

(c) Work out the total surface area of the slice. Show your working. **[2]**

Working

Answer ____ cm^2

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5 This question is about the Year 8 trip to Luxor. **[1]**

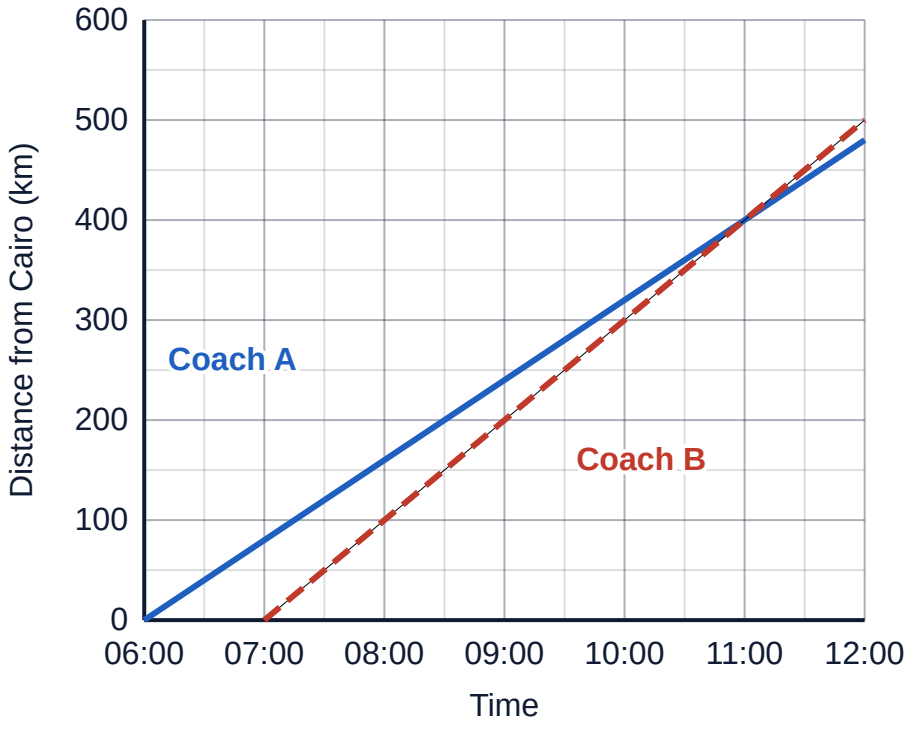
(a) The coach travels 660 km from Cairo to Luxor.
1 mile = 1.6 km. Convert 660 km to miles.

Working

Answer ____ miles

6 This question is about the Year 8 trip to Luxor. **[1]**

Two coaches leave Cairo for Luxor. The graph shows their journeys from 06:00 to 12:00.



(b)(i) At what time does Coach B pass Coach A?

(b)(ii) How far from Cairo are the coaches when this happens? **[1]**

Working

Answer ____ km

(b)(iii) Which coach travels faster? Explain how the graph shows this. **[1]**

7 This question is about the Year 8 trip to Luxor. **[1]**

A hotel in Luxor charges a booking fee of E£300 plus E£450 for each night.

(c) Write an equation for the total cost y (in E£) for x nights.

(d) The total cost for one teacher's stay is E£2550. **[1]**
How many nights did the teacher stay?

Working

Answer

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8 Here is a table of values for y = 3x − 2. **[1]**

x	−1	0	1	2	3
y	?	−2	1	?	7

(a) Work out the value of y when x = −1.

Working

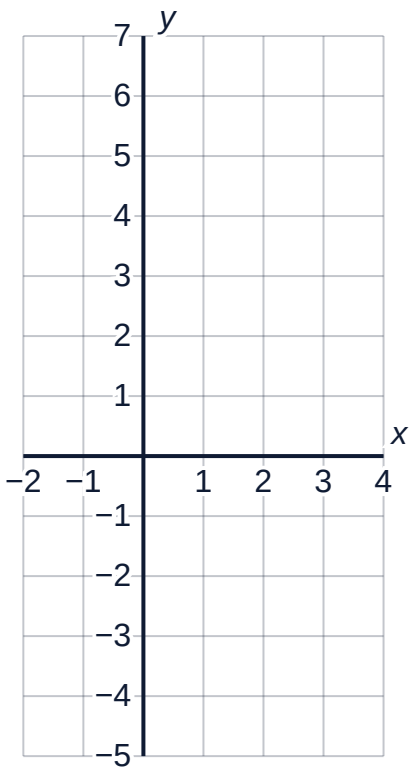
Answer

(b) Work out the value of y when x = 2. **[1]**

Working

Answer

[1]

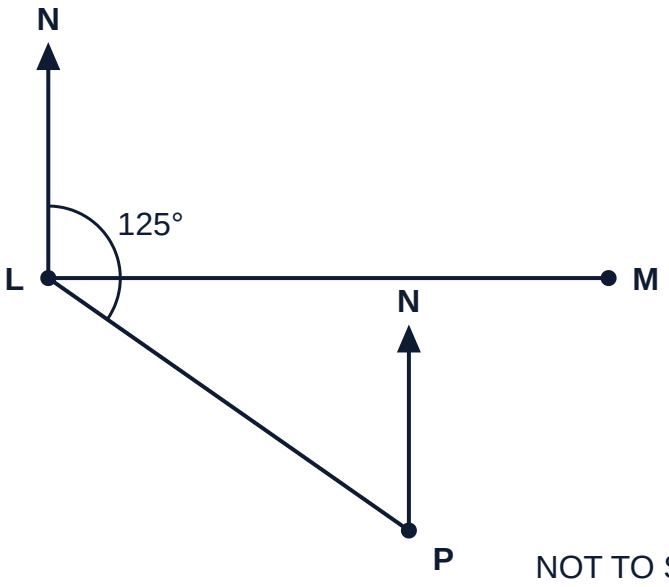


(c) Draw the graph of y = 3x − 2 for values of x from −1 to 3. (Copy the grid onto squared paper.)

(d) Which line is **parallel** to y = 3x − 2? **[1]**

- ☐ A) y = 2x − 3
- ☐ B) y = 3x + 4
- ☐ C) y = −3x − 2
- ☐ D) y = x − 2

9 The map shows three places in a city: a library L, a mosque M and a park P. **[1]**
M is due east of L. The bearing of P from L is 125°.



(a) Work out the bearing of L from P.

Working

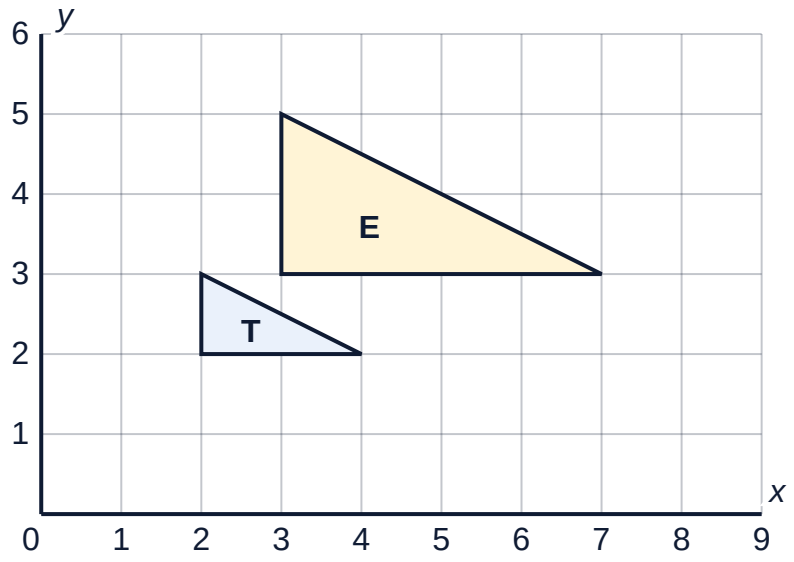
Answer ____ °

(b) Work out the size of angle MLP. **[1]**

Working

Answer ____ °

10 Triangle T has vertices (2, 2), (4, 2) and (2, 3). Triangle E is an enlargement of triangle T. **[1]**



(a) Write down the scale factor of the enlargement.

Working

Answer

(b) Find the coordinates of the centre of enlargement. **[1]**

[1]

$$\text{vector} \begin{pmatrix} 4 \\ -2 \end{pmatrix}$$

(c) Triangle T is translated by the vector shown.
Write down the coordinates of the three vertices of the image.

(d) To describe a rotation fully, which information do you need? **[1]**

- ☐ A) The centre and the angle only
- ☐ B) The angle and the direction only
- ☐ C) The centre, the angle and the direction
- ☐ D) The centre, the scale factor and the direction

11 **[1]**

(a) Ravi says, "A regular polygon with rotational symmetry of order 9 has 9 lines of symmetry and 9 sides."
Is Ravi correct? Explain your answer.

(b) A 3D shape has 12 vertices and 18 edges. **[1]**
Use Euler's formula to work out the number of faces.

Working

Answer

(c) The plan (top view) of a solid is a circle. The front elevation is a rectangle. **[1]**
Name the solid.

12 120 students visited the school science fair. Each student voted for their favourite project area. **[1]**

	Physics	**Chemistry**	**Biology**	**Total**
Boys	18	15	?	50
Girls	22	?	25	70
Total	40	?	?	120

(a) How many girls voted for Chemistry?

Working

Answer

(b) Salma draws a pie chart to show the votes of all 120 students. **[2]**
Work out the angle for Biology. Show your working.

Working

Answer ____ °

(c) For her own science project Salma collects data. Which of these is **continuous** data? **[1]**

- ☐ A) The number of students in a class
- ☐ B) Favourite science subject
- ☐ C) Shoe size
- ☐ D) The mass of a plant in grams

(d) Laila wants to know what all 600 students in the school think of the science fair. She asks the 30 students in her own science club. **[1]**
Give **one** reason why this may not be a good sample.

13 (continued) **[2]**

(e) Classes 8A and 8B took the same science quiz, marked out of 20.

Class	Mean	Range
8A	14.2	6
8B	12.8	15

Compare the scores of the two classes. Write **two** comparisons.

14 Omar has a spinner with four equal sections coloured red, blue, green and yellow. He spins it 200 times. It lands on red 62 times. **[1]**

(a) Write down the experimental probability that the spinner lands on red.

Working

Answer

(b) Omar says, "The spinner is biased towards red." **[1]**
Use the theoretical probability to comment on what Omar says.

(c) Omar spins the same spinner 500 times. Use his experimental probability to estimate how many times it lands on red. **[1]**

Working

Answer

15 **[2]**

Question 12
Solve the equation $(3x - 1)/4 = x - 2$
Show your working.